



Research Paper

Numerical simulation of transport characteristics of Li-ion battery in different discharging modes



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HIGHLIGHTS

- A 3D electrochemical-thermal coupling model of LiFePO_4 battery is developed.
- Core reaction area changes differently on both sides in constant current mode.
- Current density reduces faster for constant voltage than that for overpotential.
- Thick electrode has positive effect in maximum output power discharge mode.

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ABSTRACT

A three-dimensional electrochemical-thermal coupling model of LiFePO_4 battery was developed based on the real multi-layer structure. The model was developed to describe the discharge process of the battery in four modes: the constant current, constant overpotential, constant voltage and maximum power discharge processes. The electric conductivity of the electrode, the electric conductivity of the current collector, the ionic conductivity of the electrolyte and the diffusion coefficient of the electrolyte were considered. The results show that: (1) In the constant current discharge mode, the core electrochemical reaction area of the positive electrode moved towards the separator while that of the negative electrode moved towards the current collector due to the change of the electrolyte concentration. (2) In the constant overpotential discharge process, the change of the average chemical reaction rate and the decreasing rate of the current density increased with increasing overpotential. (3) In the constant voltage discharge process, the current density and the temperature increased when the electrode thickness was increased. (4) In the maximum power discharge process, the output power increased with increasing electrode thickness but decreased with increasing contact resistance at the same current density.

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1. Introduction

The thermal management of lithium-ion (Li-ion) battery in electric vehicles is very important for both the battery system reliability and the vehicle performance [1–3]. An effective thermal management can benefit the efficiency and the economy of the battery under various operating temperatures and conditions. Since the electrochemical reaction and the heat generation rate during the charge and discharge processes are the key to control the temperature distribution of the battery [4–6], it is meaningful to analyze the effects of the storage of charge, the working current level, the cooling intensity and the environmental temperature on the thermal behavior of the Li-ion battery.

So far, the Li-ion battery models include the simplified electrochemical model (i.e. empirical equation), the neural network model, the equivalent circuit model, the electrochemical model, the thermal model and the electrochemical-thermal coupling model. The simplified electrochemical model is simple and convenient, but the calculation accuracy is relatively low because it ignores the electrochemical reaction and the charge transfer process [7]. The neural network models are widely used in parameter identification and estimation due to their advantages in nonlinear mapping, parallel processing, and robust error-tolerance [8]. The electrochemistry model based on porous electrode theory achieves higher accuracy in predicting the electrochemistry reaction process, but the influence of the temperature on the electrochemical parameters are not considered [9]. The applicability of the thermal model is limited to some specific batteries since the heating rate completely depends on the test data [10–12].

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Nomenclature

A_s	average active area per unit volume ($\text{m}^2 \text{m}^{-3}$)	<i>Greek letters</i>	
c_l	electrolyte concentration (mol L^{-1})	α	electron transfer coefficient
c_s^{max}	the maximum Li-ion concentration in solid phase (mol m^{-3})	θ	electrode stoichiometric ratio
c_s^{surf}	the Li-ion concentration at the interface of the solid and liquid phase (mol m^{-3})	δ	thickness (m)
c_p	the specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)	λ	thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
D_s	electronic diffusion coefficient ($\text{m}^2 \text{s}^{-1}$)	ρ	density (kg m^{-3})
D_s^{eff}	effective electronic diffusion coefficient ($\text{m}^2 \text{s}^{-1}$)	σ	electrical conductivity (S m^{-1})
D_l	diffusion coefficient of Li-ion in liquid phase ($\text{m}^2 \text{s}^{-1}$)	ϕ	electrode potential (V)
D_l^{eff}	effective diffusion coefficient of Li-ion in liquid phase ($\text{m}^2 \text{s}^{-1}$)	ϕ_l	the liquid phase potential (V)
E_{cell}	battery cell voltage (V)	ϕ_s	the solid phase potential (V)
$f_{\pm}$	the average activity coefficient of electrolyte	$\phi_{s,pe}$	the positive electric potential (V)
i_0	exchange current density (A m^{-2})	η	the over potential (V)
i_f	local current density at the surface of active material (A m^{-2})	η_{SEI}	the extra over potential caused by reactant solid surface passivation layer (V)
i_l	the current density vector in liquid phase (A m^{-2})		
n	the number of electrons transferred per reaction	<i>Subscripts and superscripts</i>	
R	general gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)	cell	battery cell
r	internal resistance (Ω)	eff	effective value
S_{ohm}	source term caused by ohmic heat (W m^{-3})	i	an electrode, negative or positive
S_T	source term of heat (W m^{-3})	l	liquid phase
S_{Vs}	source term of the solid phase of energy in electrode (W m^{-3})	ne	negative electrode
S_{Vm}	source term of the liquid phase of energy in electrode (W m^{-3})	n_cu	negative electrode collector (copper)
SOC	storage of charge (%)	s	solid phase
T	temperature (K)	ohm	ohmic potential
t_+^0	the number of Li-ion transferred in electrolyte	pe	positive electrode
$U_{\text{ref}}^{\text{eff}}$	the equilibrium electrode potential (V)	p_al	positive electrode collector (aluminum)
U_{OCP}	open circuit potential of battery (V)	sep	separator
		total	total value

The electrochemical-thermal coupling model can be considered as a special thermal model since it can describe the dynamic performance of the Li-ion battery [13]. Jeon [14] simulated the charge and discharge processes of the Li-ion battery based on porous electrode theory and obtained the temperature distribution using the transient model. It was found that the temperature increase in the discharge process was higher than that in the charge process at the same discharge/charge rate. Guo and White [15] developed a pseudo two-dimensional model of the flat Li-ion batteries with higher calculation efficiency comparing with the two-dimensional and three-dimensional models. Ye et al. [16] developed a local spiral-wound Li-ion battery electro-thermal model and found that increasing the electric resistance can reduce the battery discharge voltage.

Most of the Li-ion battery models were developed for the LiMnO_4 [17–19] and LiCoO_2 [20,21] batteries in which the real multi-layer structure was not taken into account [22]. Moreover, the internal electric resistance was usually considered as a constant. However, the LiFePO_4 battery is attracting more and more attentions [23,24] due to the higher stability during the discharge process and the better performance at high temperature.

Therefore, in this study, a three-dimensional electrochemical-thermal coupling model of the LiFePO_4 battery with the real multi-layer structure was developed to investigate the discharge process. The electric conductivities of the electrode and the current collector, and the ionic conductivity and diffusion coefficient of the electrolyte were considered. Based on the proposed model, four different discharge modes of the LiFePO_4 battery were studied including the constant current discharge, the constant total over-

potential discharge, the constant voltage discharge and the constant maximum power discharge processes.

2. Numerical model

2.1. Computational domain

In general, a Li-ion battery consists of a positive current collector, a positive electrode, a separator, a negative electrode, and a negative current collector, as shown in Fig. 1(a). The 3D model (Fig. 1(b)) developed in this study was $170 \mu\text{m} \times 100 \mu\text{m} \times 200 \mu\text{m}$ along the x, y and z directions, respectively. In the x direction, the numbers of grids in different layers were the same, while those in the y and z directions varied with the model size. Based on the grid independence tests carried out in all three directions, the computational domain was meshed with 50,000 hexahedron elements in total.

2.2. Assumptions

According to the working condition and the structure characteristics of Li-ion batteries, the electrochemical-thermal coupling model was developed based on the following assumptions:

- (1) There was no gas produced in the cell in the charge and discharge processes so that only solid and liquid phases were considered;
- (2) The battery electrode volume and porosity remained constants in the charge and discharge processes;

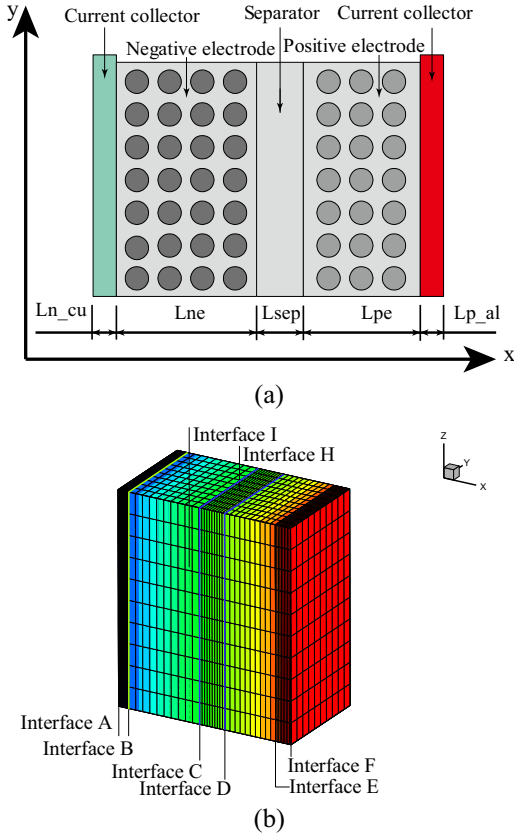


Fig. 1. Schematic (a) and mesh (b) of the LiFePO₄ battery modeled in this study (170 $\mu\text{m} \times 100 \mu\text{m} \times 200 \mu\text{m}$ along the x, y and z directions).

- (3) The pore shape and active area per unit volume in different electrode regions were the same;
- (4) The side reactions inside the battery were ignored.

2.3. Conservation equations

In the proposed model, the conservation equations of the electric and ionic potentials, the mass and the energy were solved individually for different layers of the computational domain.

- (1) Electric potential (solved in positive/negative current collector and electrode)

$$\nabla \cdot (-\sigma_s^{\text{eff}} \nabla \phi_s) = S_{Vs} \quad (1)$$

where σ_s^{eff} (S m^{-1}) is the effective conductivity of the solid electrodes, ϕ_s (V) the solid phase potential and S_{Vs} (A m^{-3}) the source term which is given as follows:

$$S_{Vs} = \begin{cases} -J & \text{in negative/positive electrode} \\ 0 & \text{in negative/positive current collector} \end{cases} \quad (2)$$

$$J = A_s i_f \quad (3)$$

where A_s ($\text{m}^2 \text{m}^{-3}$) is the average active area per unit volume, and i_f (A m^{-2}) the local current density at the active material surface:

$$i_f = i_0 \left\{ \exp \left(\frac{\alpha_a n F}{RT} (\eta - \eta_{\text{SEI}}) \right) - \exp \left(-\frac{\alpha_c n F}{RT} (\eta - \eta_{\text{SEI}}) \right) \right\} \quad (4)$$

$$i_0 = F k_0 (c_1)^{\alpha_a} (c_s^{\text{max}} - c_s^{\text{surf}})^{\alpha_a} (c_s^{\text{surf}})^{\alpha_c} \quad (5)$$

$$\eta = \phi_s - \phi_l - U_{\text{ref}}^{\text{eff}} \quad (6)$$

where i_0 (A m^{-2}) is the exchange current density, F ($96,487 \text{ C mol}^{-1}$) the Faraday constant, α_a (0.5) and α_c (0.5) the electron transfer coefficient of negative and positive electrodes respectively, R ($8.314 \text{ J mol}^{-1} \text{K}^{-1}$) the general gas constant, T (K) the temperature, and n the number of electrons transferred per reaction. η (V) is the over potential, η_{SEI} (V) the extra over potential caused by passivation layer at the reactant solid surface and it is zero there, k_0 ($\text{m}^{2.5} \text{mol}^{-0.5} \text{s}^{-1}$) the reaction rate constant, c_1 (mol m^{-3}) the Li-ion concentration in liquid phase, c_s^{max} (mol m^{-3}) the maximum Li-ion concentration in solid phase, c_s^{surf} (mol m^{-3}) the Li-ion concentration at the interface of the solid and liquid phase, i.e. the surface of the reaction material particles, ϕ_l (V) the liquid phase potential and $U_{\text{ref}}^{\text{eff}}$ (V) the equilibrium electrode potential.

- (2) Ionic potential (solved in positive/negative electrode and separator)

$$\nabla \cdot \mathbf{i}_l = S_{Vm} \quad (7)$$

where i_l (A m^{-2}) is the current density vector in liquid phase, and S_{Vm} (A m^{-3}) the source term:

$$\mathbf{i}_l = -\sigma_l^{\text{eff}} \nabla \phi_l + \frac{2RT\sigma_l^{\text{eff}}}{F} \left(1 + \frac{\partial \ln f_{\pm}}{\partial \ln c_1} \right) (1 - t_+^0) \nabla (\ln c_1) \quad (8)$$

$$S_{Vm} = \begin{cases} J & \text{in negative/positive electrode} \\ 0 & \text{in separator} \end{cases} \quad (9)$$

where σ_l^{eff} ($\text{m}^2 \text{s}^{-1}$) is the effective conductivity in liquid electrodes, $f_{\pm}$ the average activity coefficient of electrolyte, and t_+^0 (0.363) the number of Li-ion transferred in electrolyte. Note that the thermodynamic factor depends on the temperature and the concentration, which can be expressed as [25]:

$$\begin{aligned} & \left(1 + \frac{\partial \ln f_{\pm}}{\partial \ln c_1} \right) (1 - t_+^0) \\ &= \frac{1 + t_+^0}{1 - 0.399} \left(0.601 + 0.24 \times 0.01 c_1^{0.5} + 0.982 \right. \\ & \quad \times \left[1 - 0.0052(T - 294) \right] (0.001 c_1)^{1.5} \Big) \end{aligned} \quad (10)$$

- (3) Mass conservation (solved in positive/negative electrode and separator)

$$\frac{\partial}{\partial t} (\varepsilon_l c_1) + \nabla \cdot (-D_l^{\text{eff}} \nabla c_1) = \frac{J}{F} - \nabla \cdot \left(\frac{t_+^0 \mathbf{i}_l}{F} \right) \quad (11)$$

where D_l^{eff} ($\text{m}^2 \text{s}^{-1}$) is the effective diffusion coefficient of Li-ion in liquid phase and ε_l (%) the volume fraction of the liquid phase.

- (4) Energy (solved in all the computational domain)

$$\frac{\partial}{\partial t} (\rho c_p T) + \nabla \cdot (-\lambda \nabla T) = S_T \quad (12)$$

where ρ (kg m^{-3}) is the density, c_p ($\text{J kg}^{-1} \text{K}^{-1}$) the specific heat capacity, λ ($\text{W m}^{-1} \text{K}^{-1}$) the thermal conductivity and S_T (W m^{-3}) the source term shown as follows:

$$S_T = \begin{cases} J\eta + JT \frac{\partial U_{\text{ref}}}{\partial T} + S_{\text{ohm}} & \text{in negative/positive electrode} \\ S_{\text{ohm}} & \text{others} \end{cases} \quad (13)$$

where the entropic change can be expressed as [26]:

$$\frac{\partial U_{\text{ref,ne}}}{\partial T} = 344.1348147 \times \frac{\exp(-32.9633287\theta_{\text{ne}} + 8.316711484)}{1 + 749.0756003 \exp(-34.79099646\theta_{\text{ne}} + 8.887143624)} - 0.8520278805\theta_{\text{ne}} + 0.362299229\theta_{\text{ne}}^2 + 0.2698001697 \quad (14)$$

$$\frac{\partial U_{\text{ref,pe}}}{\partial T} = -0.35376\theta_{\text{pe}}^8 + 1.3902\theta_{\text{pe}}^7 - 2.2585\theta_{\text{pe}}^6 + 1.9635\theta_{\text{pe}}^5 - 0.98716\theta_{\text{pe}}^4 + 0.28857\theta_{\text{pe}}^3 - 0.046272\theta_{\text{pe}}^2 + 0.0032158\theta_{\text{pe}} - 1.9186 \times 10^{-5} \quad (15)$$

where the open circuit potential of the positive/negative electrode can be expressed as [27]:

$$U_{\text{ref,pe}} = 3.4323 - 0.4528 \exp(-80.2493(1 - \theta_{\text{pe}})^{1.3198}) - 3.2474 \times 10^{-6} \exp(20.2645(1 - \theta_{\text{pe}})^{3.8003}) + 3.2482 \times 10^{-6} \exp(20.2646(1 - \theta_{\text{pe}})^{3.7995}) \quad (16)$$

$$U_{\text{ref,ne}} = 0.6379 + 0.5416 \exp(-305.5309\theta_{\text{ne}}) + 0.044 \tanh\left(-\frac{\theta_{\text{ne}} - 0.1958}{0.1088}\right) - 0.1978 \tanh\left(\frac{\theta_{\text{ne}} - 1.0571}{0.0854}\right) - 0.6875 \tanh\left(\frac{\theta_{\text{ne}} - 0.0117}{0.0529}\right) - 0.0175 \tanh\left(\frac{\theta_{\text{ne}} - 0.5692}{0.0875}\right) \quad (17)$$

And the source term caused by ohmic heat is:

$$S_{\text{ohm}} = \begin{cases} \sigma_s^{\text{eff}}(\nabla \phi_s)^2 + \sigma_l^{\text{eff}}(\nabla \phi_l)^2 + \frac{2RT\sigma^{\text{eff}}}{F}(1 - t_+^0)\nabla(\ln c_l) \cdot \nabla \phi_l & \text{in ne, pe} \\ \sigma_l^{\text{eff}}(\nabla \phi_l)^2 + \frac{2RT\sigma^{\text{eff}}}{F}(1 - t_+^0)\nabla(\ln c_l) \cdot \nabla \phi_l & \text{in sep} \\ \sigma_s^{\text{eff}}(\nabla \phi_s)^2 & \text{in n.cu, p.al} \end{cases} \quad (18)$$

2.4. Initial and boundary conditions

Fig. 2 shows two sorts of boundary conditions [28] and the Method 2 was adopted in this study.

For interface A:

$$\phi_s|_A = \eta_{\text{total}} = U_{\text{OCP}} - E_{\text{cell}} \quad (19)$$

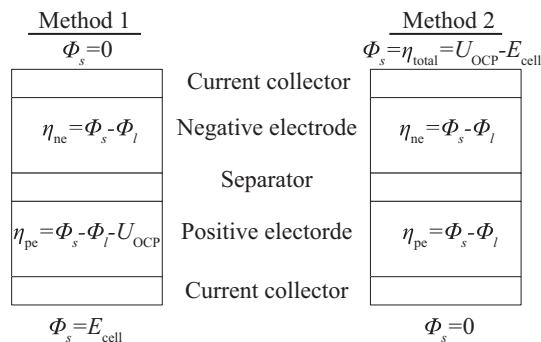


Fig. 2. Two different electric/ionic potential boundary conditions and corresponding overpotentials in negative and positive electrode. [28].

For interface F:

$$\phi_s|_F = 0 \quad n \cdot q|_{F+} = n \cdot q|_{F-} \quad (20)$$

In terms of the thermal boundary condition, the constant heat convection boundary condition was defined at the interface F with the convective heat transfer coefficient of $1 \text{ W m}^{-2} \text{ K}^{-1}$, and all the other boundaries were adiabatic.

The initial conditions of the model were:

$$c_s = c_{s,0} \quad c_l = c_{l,0} \quad T = T_0 \quad \phi_{s,\text{pe}} = \phi_s|_F = 0 \quad \phi_{l,\text{ne}} = \phi_{l,\text{sep}} = \phi_{l,\text{pe}} = 0 \quad (21)$$

where $\phi_{s,\text{pe}}$ (V) is the positive electric potential, and E_{cell} (V) the cell voltage. The subscript pe represents the positive electrode, ne the negative electrode and sep the separator.

2.5. Numerical procedures

The transport parameters used in the conservation equations are shown in Table 1, and the parameters utilized in the LiFePO₄ battery model are shown in Table 2 [26,29–31]. The conservation equations were implemented in the FLUENT software with the help of user-defined function (UDF), and the convergence criteria with a residual of 10^{-10} was used for all the variables in this study.

2.6. Model validation

A commercial LiFePO₄ battery with capacity of 10.5Ah was tested using the NBT (BTS5300C4 N, China, 0–5 V, 0–300 A) test system, and the charge–discharge current and battery voltage were recorded. The battery was placed in a temperature controlled box ($-40^\circ\text{C} \sim 150^\circ\text{C}$) under 25°C environment temperature and 100% SOC. For the charge process, the battery was firstly charged with a constant current of 3.5 A until the battery voltage achieved 3.65 V, then switched to a constant voltage charge until the current dropped to 550 mA. For the discharge process, the battery was discharged at different constant currents of 1 C, 2 C, and 5 C at room temperature of 25°C .

In order to validate the numerical model, the voltage change and the discharge time during the discharge process at different

Table 1
Transport parameters.

Parameter	Expression
Electrode stoichiometric ratio	$\theta = \frac{c_s^{\text{surf}}}{c_s^{\text{max}}}$
Storage of charge (%)	$\text{SOC} = 100 \frac{\theta - \theta_0}{\theta_1 - \theta_0}$
Electronic conductivity (S m^{-1})	$\sigma_s^{\text{eff}} = \sigma_s c_s^\gamma$
Ionic conductivity (S m^{-1})	$\sigma_l^{\text{eff}} = \sigma_l c_l^\gamma$
Electronic diffusion coefficient ($\text{m}^2 \text{ s}^{-1}$)	$D_s^{\text{eff}} = D_s c_s^\gamma$
Ion diffusion coefficient ($\text{m}^2 \text{ s}^{-1}$)	$D_l^{\text{eff}} = D_l c_l^\gamma$
Li-ion conductivity in the electrolyte (S m^{-1}) [25]	$\sigma_l = 10^{-4} c_l \times 1.2544 \times (-8.2488 + 0.532487 - 0.00002987T^2 + 0.26235 \times 0.001 c_l - 0.0093063 \times 0.001 c_l T + 0.00008069 \times 0.001 c_l T^2 + 0.22002 \times 10^{-6} c_l^2 - 0.0001765 \times 10^{-6} c_l^2 T)^2$
Lithium ion diffusion coefficient in the electrolyte ($\text{m}^2 \text{ s}^{-1}$) [25]	$D_l = 10^{(-4.43 - \frac{E_{\text{cell}}}{T - 229 - 0.005 c_l - 0.00022 c_l}) - 4}$
Physical and chemical parameters	$\phi = \phi_{\text{ref}} \exp\left[\frac{E_{\text{cell}}}{R} \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T}\right)\right]$
Open circuit potential (V)	$U_{\text{OCP}} = U_{\text{ref,pe}}^{\text{eff}} - U_{\text{ref,ne}}^{\text{eff}}$
Electrode potential (V)	$U_{\text{ref,i}}^{\text{eff}} = U_{\text{ref,i}} + (T - T_{\text{ref}}) \frac{\partial U_{\text{ref,i}}}{\partial T} \quad i = \text{ne, pe}$

Table 2
LiFePO₄ battery model parameters.

Parameter	Copper	Negative electrode	Separator	Positive electrode	Aluminum
$L(\text{m})$ [E]	10×10^{-6}	142×10^{-6}	25×10^{-6}	91×10^{-6}	15×10^{-6}
ε_l [29]	–	0.332	0.54	0.33	–
ε_p [M]	–	0.14	–	0.19	–
ε_s [29]	–	0.43	–	0.55	–
θ_0^{E}	–	0.006	–	0.993	–
θ_1^{E}	–	0.878	–	0.216	–
c_l (mol m ⁻³)	–	1200 [29]	–	–	–
D_s (m ² s ⁻¹)	–	1.18×10^{-18} [30]	–	3.9×10^{-14} [29]	–
σ_s (S m ⁻¹) [31]	6.0×10^7	100	–	3.8	3.8×10^7
k_0 (m ^{2.5} mol ^{-0.5} s ⁻¹)	–	1.4×10^{-12} [29]	–	3×10^{-11} [29]	–
E_{a,D_s} (kJ mol ⁻¹) [31]	–	4	–	20	–
E_{a,D_l} (kJ mol ⁻¹) [31]	–	10	10	10	–
E_{a,σ_s} (kJ mol ⁻¹) [E]	–	0	–	0	–
E_{a,σ_l} (kJ mol ⁻¹) [E]	–	10	10	10	–
γ [29]	–	1.5	–	–	–
T_{ref} (K)	298.15	–	–	–	–
ρ (kg m ⁻³) [26]	8700	2223	900	1500	2700
c_p (J kg ⁻¹ K ⁻¹) [26]	396	641	1883	800	897
λ (W m ⁻¹ K ⁻¹) [26]	398	1.04	0.5	1.48	237

In this table, [M] data is from OEM, [E] data is estimated.

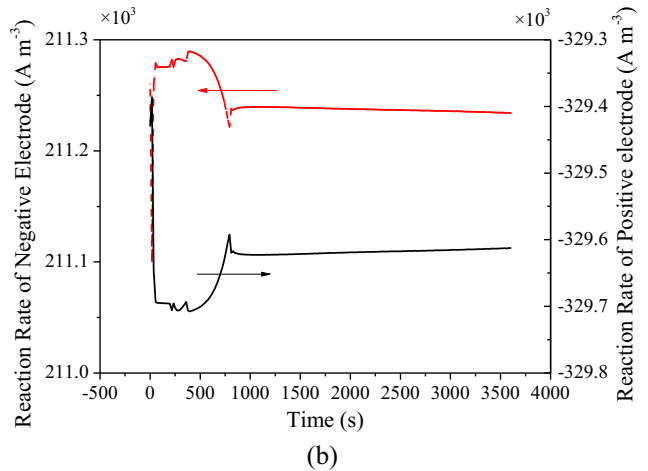
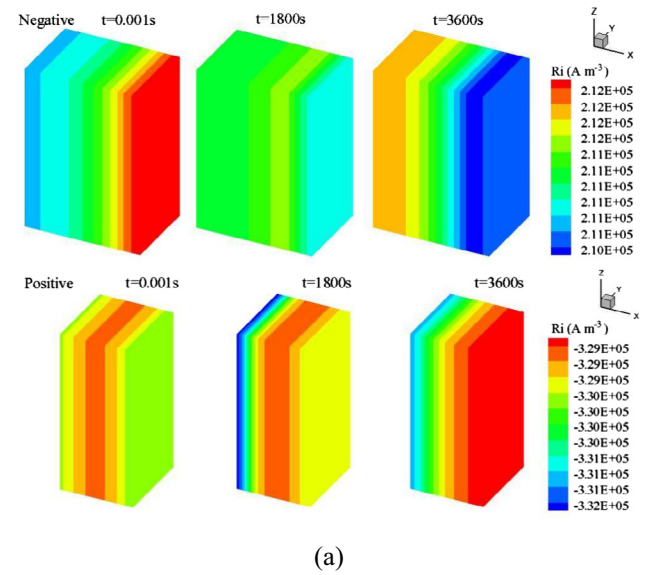
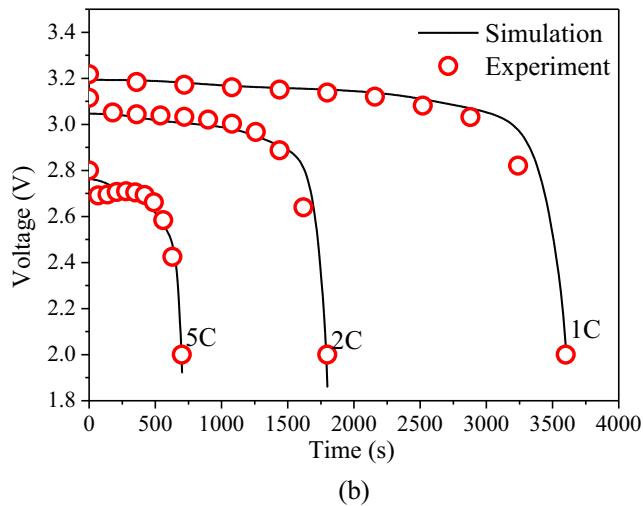
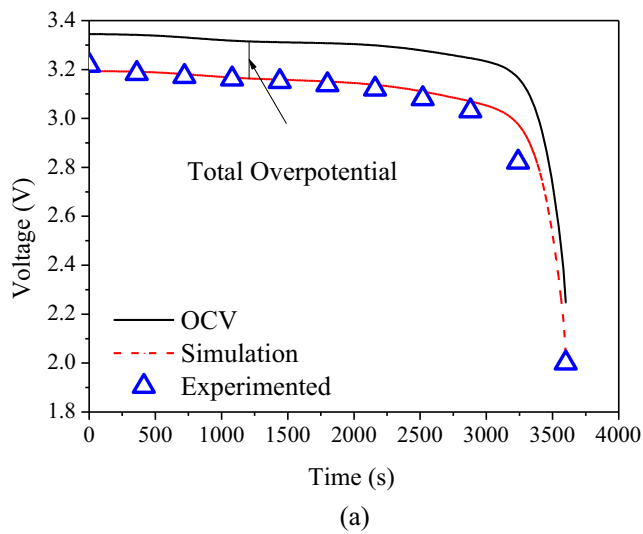


Fig. 3. Modeled battery discharge voltage and experimental value: (a) The difference between 1 C discharge voltage and experimental value; (b) Comparison of different discharge rate (1 C, 2 C and 5 C) voltage and experimental value.

Fig. 4. Electrochemical reaction rate in 1 C constant current discharge process: (a) Distribution of positive and negative electrochemical reaction rate at different discharge time; (b) The average change of the electrochemical reaction rate of positive and negative electrode.

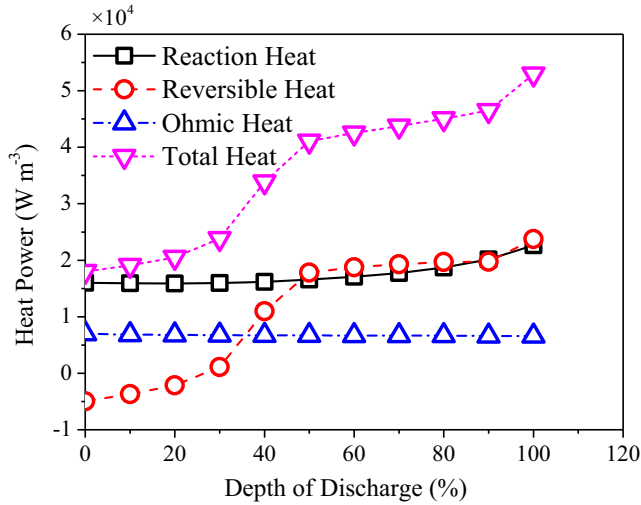


Fig. 5. Reaction, reversible, ohmic and total heat generation rate of the battery at 1 C discharge rate.

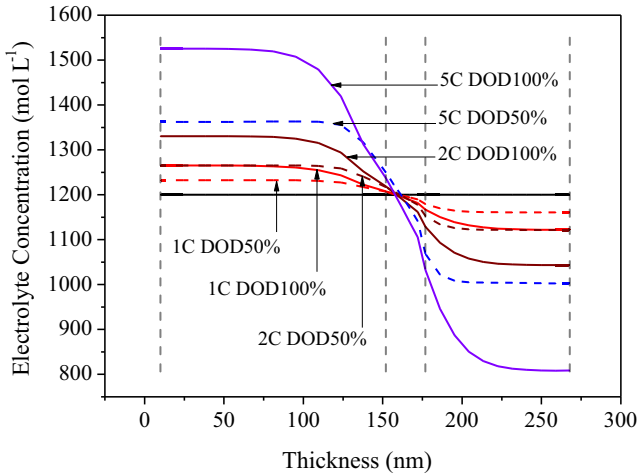


Fig. 6. Distribution of electrolyte concentration at different discharge rate (1 C, 2 C and 5 C).

rates were simulated and compared with the experimental results, as shown in Fig. 3. It can be observed in Fig. 3(a) that the battery voltage calculated by the model agrees with the tested data well at 1 C discharge rate. Note that the difference between the open circuit voltage and the operating voltage is the total overpotential.

Moreover, in Fig. 3(b), the simulation results at 1 C, 2 C and 5 C discharge rate are all consistent with the corresponding experiment results.

3. Results and discussion

In this study, the change of the electrochemical reaction rate distribution in electrode, the heat generation rate and the effect of discharge rate in the constant current discharge mode, the constant total overpotential discharge mode, the constant voltage discharge mode, and the constant maximum power discharge mode were investigated in detail.

3.1. Constant current discharge

Fig. 4 (a) shows the electrochemical reaction rate distribution of the positive and negative electrodes at 1 C discharge rate. At the beginning ($t = 0.001$ s), the reaction rates of both the negative and positive electrodes were evenly distributed. At $t = 1800$ s, the core reaction region of the positive electrode shifted towards the separator side because the electrolyte concentration close to the positive collector decreased rapidly, while that of the negative electrode moved towards the negative collector due to the rapid increase of the electrolyte concentration near the negative collector. Moreover, it should be noticed in Fig. 4(b) that the absolute values of the average reaction rates at both the negative and positive electrodes increased quickly from 0 to around 750 s, and then decreased slightly in a very short time period to a relatively stable level.

The heat generation rate at 1 C discharge rate is shown in Fig. 5. In the whole discharge process, the ohmic heat decreased slightly due to the reduction of the internal electric resistance. The reversible heat was negative at the beginning, became positive when the depth of discharge (DOD) reached 35%, increased sharply until 50% DOD, kept stable from 50 to 90% DOD, and increased slightly in the final stage. The reaction heat took a large proportion of the total heat at the beginning, but it only increased a little in the final discharge stage.

In addition, the effect of the discharge rate on the electrolyte concentration was also investigated and the results are shown in Fig. 6. The initial electrolyte concentrations of different discharge rates were the same. With changing discharge rate, the electrolyte concentration changes accordingly. In the discharge process, the electrolyte concentration in the negative region increased while that in the positive region decreased. Meanwhile, the electrolyte concentration in the positive region was higher than that in negative region due to the thinner electrode and higher reaction rate in the positive region.

Table 3

Positive and negative overpotentials at different DOD under different discharge rates.

DOD (%)	1 C discharge		2 C discharge		5 C discharge	
	$\eta_{ne}(V)$	$\eta_{pe}(V)$	$\eta_{ne}(V)$	$\eta_{pe}(V)$	$\eta_{ne}(V)$	$\eta_{pe}(V)$
0	0.064951	-0.08627	0.097463	-0.12056	0.143565	-0.16725
10	0.063704	-0.08677	0.096664	-0.12215	0.144116	-0.17145
20	0.062822	-0.08729	0.096201	-0.12373	0.145177	-0.17592
30	0.062604	-0.08817	0.096398	-0.12559	0.147136	-0.18101
40	0.063197	-0.08969	0.097575	-0.12819	0.150629	-0.18754
50	0.064624	-0.09187	0.099878	-0.13173	0.156189	-0.19617
60	0.066783	-0.09448	0.103069	-0.13577	0.163747	-0.20669
70	0.069957	-0.09769	0.107386	-0.14041	0.173973	-0.21963
80	0.074787	-0.10186	0.113541	-0.14603	0.188551	-0.23642
90	0.082646	-0.10758	0.123059	-0.1533	0.21112	-0.25974
100	0.097825	-0.11599	0.140623	-0.16353	0.252157	-0.29561

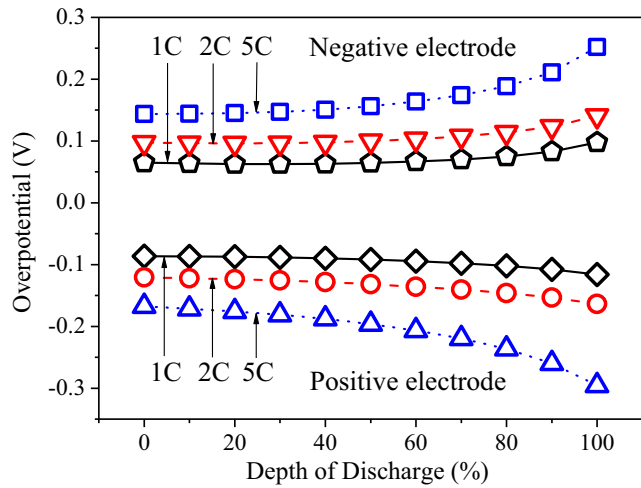


Fig. 7. Change of overpotential in negative and positive electrode at different discharge rate (1 C, 2 C and 5 C).

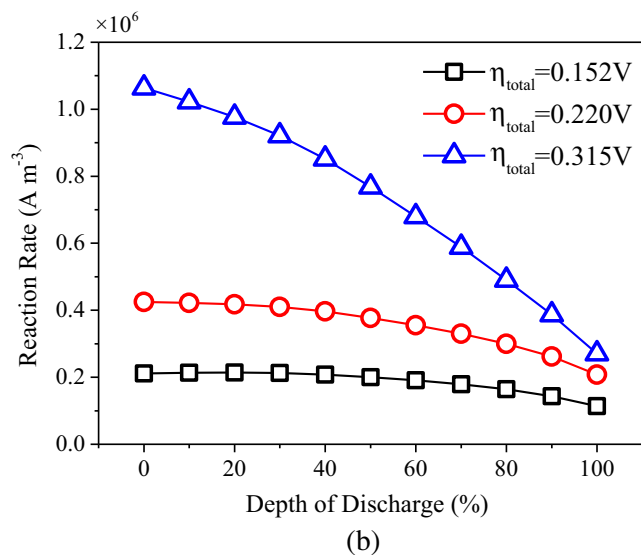
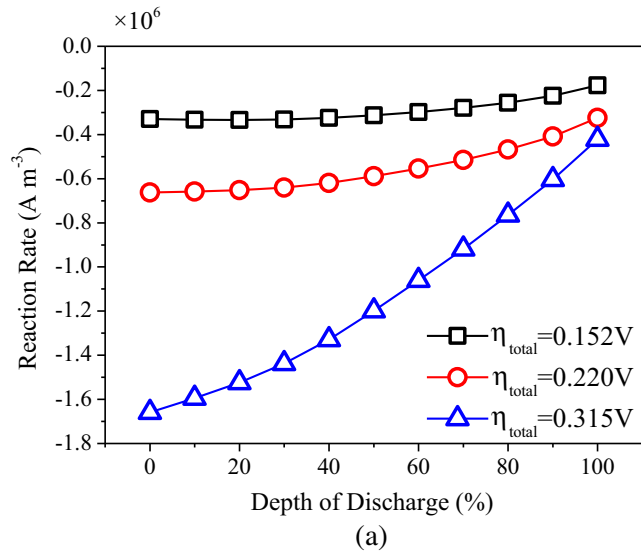


Fig. 8. The average chemical reaction rate of positive and negative electrode under three different total overpotential conditions: (a) Positive average chemical reaction rate; (b) Negative average chemical reaction rate.

The average overpotential of both the positive and negative electrodes at different discharge rates and DODs are shown in Table 3 and Fig. 7. Obviously, the overpotential increased with the increment of DOD monotonously, and the increment of the positive overpotential became larger as the discharge rate raised. When the DOD increased, the overpotential in the negative electrode at 1 C and 2 C discharge rates first decreased slightly and then increased, But under the 5 C condition, the negative overpotential increased with increasing DOD monotonously.

3.2. Constant overpotential discharge

Under a constant total overpotential, the discharge current density and the chemical reaction rate decrease with the increase of DOD. In this section, parameters regarding the electrodes and the electrochemistry were studied under three overpotential conditions: 0.152 V, 0.220 V, and 0.315 V. Fig. 8 shows that the reaction rate of both positive and negative electrodes decreased with the increment of DOD in the constant total overpotential discharge mode. Moreover, the larger the total overpotential, the larger the reaction rate, but the difference decreased with

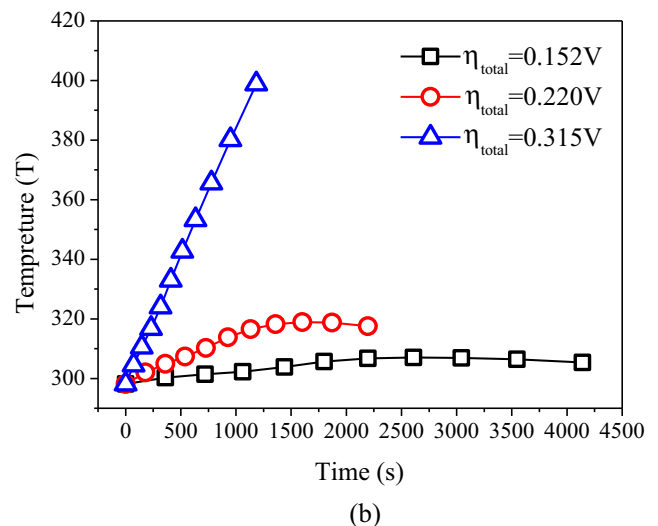
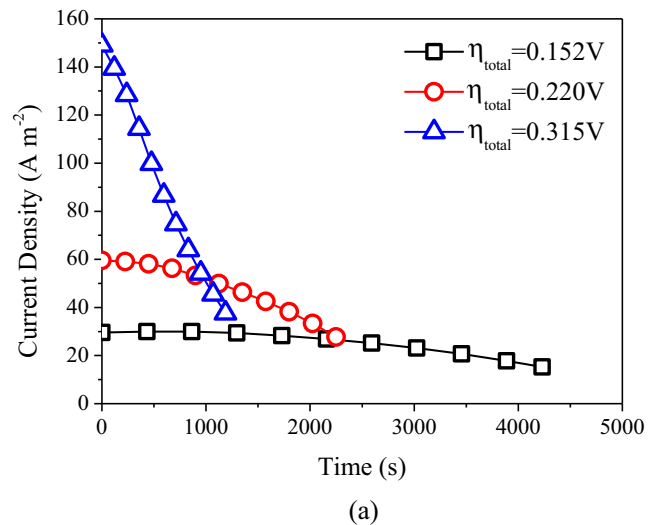


Fig. 9. Evolutions of the current density and the temperature under various total overpotentials: (a) Current density; (b) temperature.

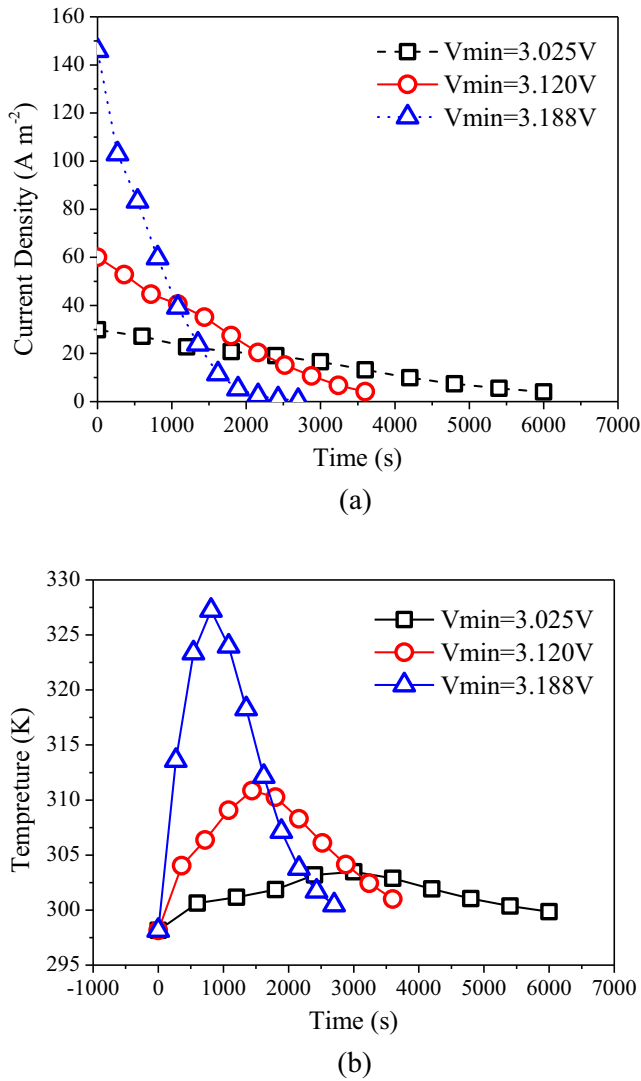


Fig. 10. The changes of electrochemical parameters during constant voltage discharge of battery: (a) Current density; (b) temperature.

the increment of DOD (at the beginning, the maximum reaction rate was five times of the minimum, but it decreased to about twice finally).

Fig. 9 illustrates how the discharge current density and temperature vary with the time. In Fig. 9(a), it is evident that the larger the overpotential, the larger the initial current density and the decreasing rate. Moreover, it should be noticed that the decreasing rate increased under total overpotentials of 0.152 V and 0.22 V, but it increased first and then decreased under total overpotential of 0.315 V. Meanwhile, it can be observed in Fig. 9(b) that the larger the total overpotential, the larger the maximum temperature and the increasing rate. In the cases of 0.152 V and 0.22 V total

overpotentials, the temperature increased first and then decreased; while it increased monotonically in the case of 0.315 V total overpotential.

3.3. Constant voltage discharge

The constant voltage discharge mode was simulated with the same initial current density as the constant total overpotential case so as to compare the current density and temperature variations in the discharge process. The voltages corresponding to the total overpotential of 0.152 V, 0.22 V and 0.315 V were calculated as 3.188 V, 3.12 V and 3.025 V respectively. Fig. 10 shows the current density and temperature under the constant voltage discharge mode. It can be seen in Fig. 10(a) that the current density decreased to zero in the discharge process, and the decreasing rate also decreased and finally remained stable. Moreover, the decreasing rate of the current density was larger under the constant voltage discharge mode than that under the constant total overpotential. However, in the constant total overpotential case, the discharge time was longer and consistent state at the end of the discharge was better.

Meanwhile, Fig. 10(b) shows that the temperature increased to a peak and then decreased under the constant voltage discharge mode. The temperature changing rate and the maximum temperature increased with the decreasing voltage. However, it must be noticed that the electricity energy can be released completely under the constant overpotential discharge mode, but it cannot under constant voltage discharge mode because that the voltage is higher than the minimum battery voltage.

3.4. Constant maximum power discharge

Further to the study in Section 3.3, the extreme condition of the constant voltage discharge, the maximum power discharge, was studied. In theory, the ideal maximum power discharge can be achieved at the minimum discharge voltage, but the maximum power calculated in this numerical model was lower than the ideal value due to the electric resistance between the electrode tab and the plate (defined as the contact resistance). Three different batteries with various positive and negative electrode thicknesses were considered in this section, as shown in Table 4.

Fig. 11 shows the evolutions of the current density and the temperature under different minimum discharge voltages. It can be concluded that both the current density and the temperature increased with the increment of electrode thickness due to the increase of heat generation rate. However, the decreasing trend of the current density with the increase of DOD is the same regardless of the electrode thickness.

Moreover, the evolutions of cell voltage and discharge power of three kinds of batteries with different thicknesses of positive and negative electrodes under different contact resistance are shown in Fig. 12. It can be observed that at thin electrode, the output voltage decreased with the increase of contact resistance, the larger the discharge current density, the more evident the decrease trend of voltage, meanwhile the output power decreased sharply with the increment of contact resistance; with the increase of electrode

Table 4
Three kinds of batteries with different thicknesses of positive and negative electrodes.

Cell number	L_{n_cu} (nm)	L_{ne} (nm)	L_{sep} (nm)	L_{pe} (nm)	L_{p_al} (nm)
1#	10	50	25	36.4	15
2#	10	70	25	50	15
3#	10	142	25	91	15

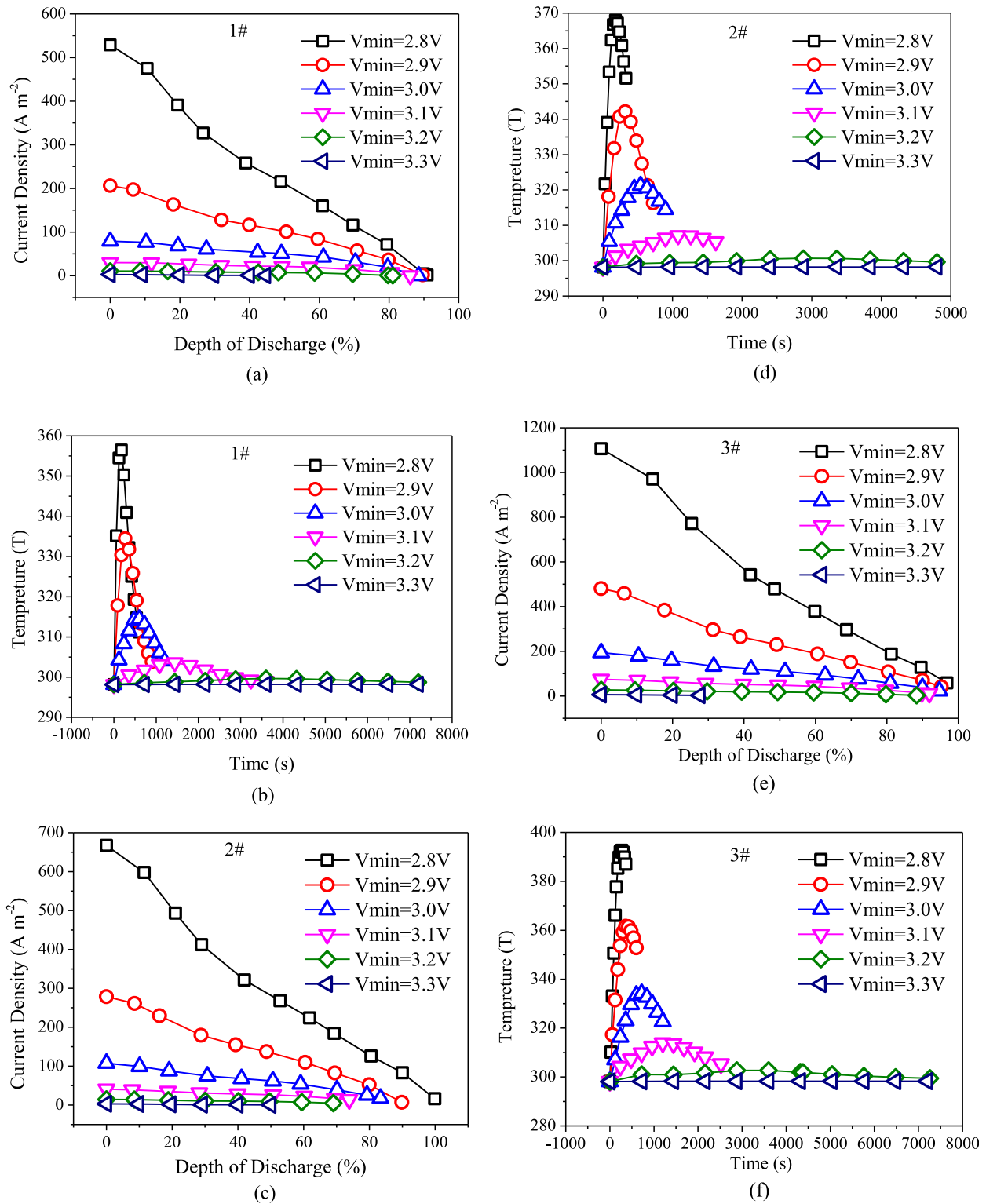


Fig. 11. Evolutions of current density and temperature of three kinds of batteries with different thicknesses of positive and negative electrodes (seen in Table 4) under different minimum discharge voltage: (a) Current density of 1# battery; (b) Temperature of 1# battery; (c) Current density of 2# battery; (d) Temperature of 2# battery; (e) Current density of 3# battery; (f) Temperature of 3# battery.

thickness, the decreasing trend of the voltage became more evident and there was an inflection point at higher contact resistance; if the electrode was thick enough, the output voltage and the power would become negative at a large enough contact resistance (e.g. 8 mΩ), in which case the theoretical current density could not be achieved.

4. Conclusion

In this study, an electrochemical-thermal coupling model was developed for the microstructure of the electrochemical reaction area of a battery. The discharge characteristics were investigated in four modes: the constant current, constant overpotential,

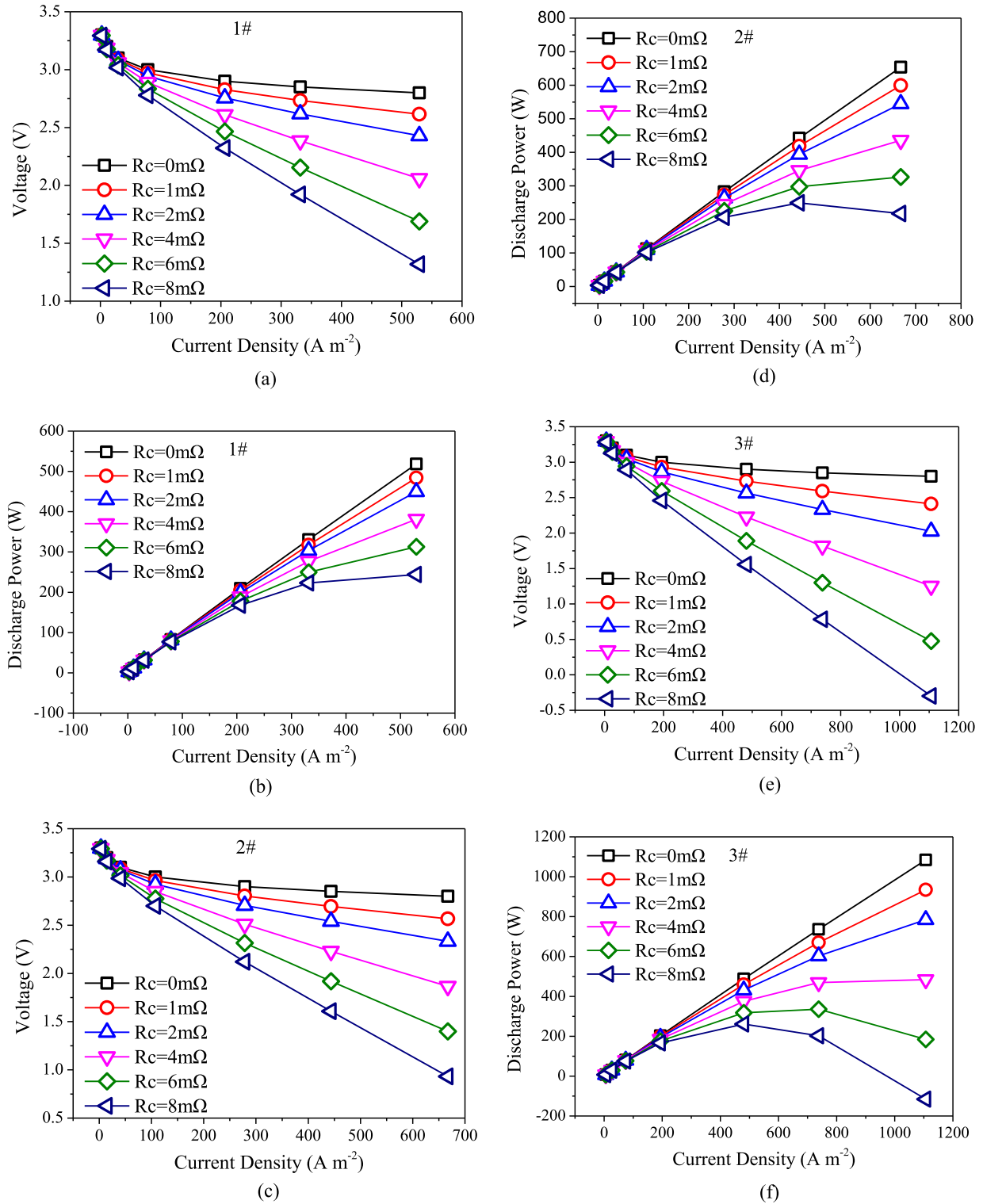


Fig. 12. Evolutions of cell voltage and discharge power of three kinds of batteries with different thicknesses of positive and negative electrodes under different contact resistance (seen in Table 4): (a) Cell voltage of 1# battery; (b) Discharge power of 1# battery; (c) Cell voltage of 2# battery; (d) Discharge power of 2# battery; (e) Cell voltage of 3# battery; (f) Discharge power of 3# battery.

constant voltage and maximum power discharge processes. For the constant current discharge process, the electrochemical reaction area of the positive electrode moved towards the separator while that of the negative electrode moved to the negative current collector due to the change of the electrolyte concentration in the discharge process of battery. Under 1 C and 2 C discharge rates, the

electrode overpotential of the negative electrode first decreased slightly and then increased. However, in the case of high rate discharge (5 C), the negative electrode overpotential increased monotonically, which means that the discharge rate can affect the change of the negative overpotential. In the constant total overpotential working condition, the current density decreased with the

discharge time, and the decreasing rate increased with increasing total overpotential. The larger the total overpotential is, the larger the maximum temperature and the increasing rate becomes. In terms of the constant discharge voltage discharge process, the battery current density reduced faster than that in the constant overpotential discharge case, but its internal consistency of the battery was better at the end of the discharge. At the same time, the temperature of the constant voltage discharge cell was smaller, and the trend of the temperature drop was more obvious. In the maximum discharge power working condition, under the same discharge voltage platform, the current density increased with the increment of electrode thickness. The greater the electrode thickness became, the higher the temperature extreme value was, and vice versa. Considering the contact resistance of the battery, the decline in the battery voltage increased as the thickness of the electrode increased, and the battery discharge power showed a knee point with the increase of current density.

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